PRACTICALS

Computer Networks Laboratory

Practical 1.

Introduction to Cisco Packet Tracer

1. Introduction

Cisco Packet Tracer is a network simulation and visualization tool developed by Cisco Systems and distributed mainly through the Cisco Networking Academy. It allows students, teachers and network enthusiasts to design, configure, test and troubleshoot computer networks on a computer without owning any physical routers, switches or cables.

Real networking hardware is expensive, bulky and difficult to provide to every learner in a laboratory. Packet Tracer solves this problem by creating a virtual environment in which devices such as routers, switches, PCs, servers, wireless access points and even IoT devices behave in a way that closely matches real Cisco equipment. A learner drags devices onto a workspace, connects them with the proper cables, configures them through a graphical interface or the Cisco IOS command line, and then watches data packets move across the network.

The tool is widely used while preparing for Cisco certifications such as CCNA and in academic courses on Computer Networks. Because it runs on Windows, macOS and Linux (and has companion mobile versions), a student can practise networking at any time. It is a simulator, not an emulator: it imitates the behaviour of devices and a selected set of protocols rather than running the actual IOS images.

Aim of this case study

To understand what Cisco Packet Tracer is, what it offers, how its interface is organised, where it is useful and where it falls short.

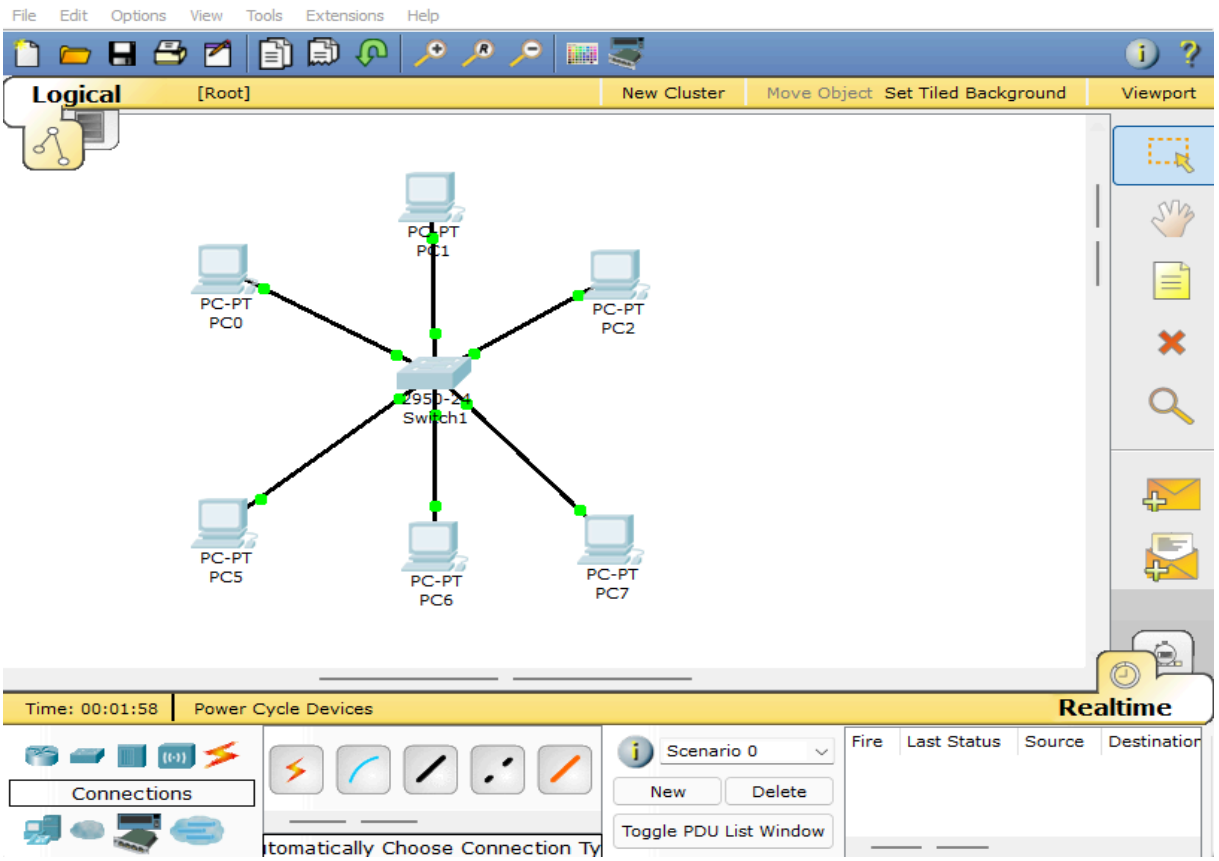
2. Key Features of Packet Tracer

- **Logical and Physical workspaces.** The Logical view is used to place devices and draw the topology, while the Physical view shows devices in a scaled model of rooms, buildings and cities.
- **Realtime and Simulation modes.** In Realtime mode the network behaves like a live network. In Simulation mode time is controlled step by step, so each packet can be tracked hop by hop.

- **Wide range of devices.** Routers, switches, hubs, wireless access points, firewalls, servers, PCs, laptops, IP phones, smartphones, and IoT things such as sensors, sirens and smart home devices.
- **Multiple configuration methods.** Devices can be configured through the Config tab (GUI), the CLI tab (Cisco IOS commands) and the Desktop tab on end devices (IP Configuration, Command Prompt, Web Browser).
- **Protocol support.** Supports commonly taught protocols and services such as IPv4, IPv6, TCP, UDP, ARP, ICMP, DHCP, DNS, HTTP, HTTPS, FTP, SMTP, Telnet, SSH, RIP, OSPF, EIGRP, BGP, VLAN, STP, NAT and ACLs.
- **Protocol Data Unit (PDU) tools.** Simple and Complex PDUs can be placed on the workspace to test connectivity (similar to ping) and to inspect packet contents at every OSI layer.
- **Multiuser mode.** Several instances of Packet Tracer can be connected over a network so that users can collaborate or compete on the same activity.
- **Activity Wizard and assessment.** Instructors can create activities with instructions, answer networks and automatic scoring.
- **Programmability.** Supports scripting in languages such as Python and JavaScript, and visual programming, which is useful for IoT and network automation exercises.
- **Save and share.** Projects are saved as .pkt files and can be shared, reopened and evaluated later.

3. Understanding the Interface

The Packet Tracer window is divided into clearly defined areas. Knowing the purpose of each one makes building a network much faster. Figure 1 shows a schematic layout of the main window.



[FIG 1: Schematic layout of the Packet Tracer main window]

Component	Purpose
Menu Bar	Contains File, Edit, Options, View, Tools, Extensions, Window and Help menus for opening, saving, preferences and activity tools.
Main Toolbar	Shortcut icons for New, Open, Save, Print, Copy, Paste, Undo, Redo, Zoom and the Network Information window.
Logical / Physical Tabs	Switch between the logical topology view and the physical (building and city) view.
Workspace	The central area where devices are placed, cabled and arranged to form the topology.
Common Tools Bar	Right-hand bar with Select, Move Layout, Place Note, Delete, Inspect, Resize Shape, Add Simple PDU and Add Complex PDU.
Device-Type Selection Box	Bottom-left box listing categories: Network Devices, End Devices, Components, Connections, Miscellaneous and Multiuser Connection.
Device-Specific Selection Box	Shows the individual models or cable types of the chosen category, such as different routers, switches or connections.
Realtime / Simulation Toggle	Bottom-right tabs to change between live operation and step-by-step packet tracing.

Component	Purpose
User Created Packet Window	Lists the PDUs created by the user and their scenarios for repeated testing.
Simulation Panel (Event List)	Appears in Simulation mode and records every packet event with time, device, protocol and layer details.

Device configuration window

Double-clicking any device opens its configuration window, which normally has the following tabs:

- **Physical:** add or remove modules and power the device on or off.
- **Config:** graphical settings such as hostname, interfaces, IP address, VLANs and routing.
- **CLI:** Cisco IOS command line for typing real configuration commands.
- **Desktop (end devices):** IP Configuration, Terminal, Command Prompt, Web Browser and other tools.
- **Attributes:** additional properties of the device.

4. Advantages and Limitations

Advantages	Limitations
Free for Cisco Networking Academy learners, so no hardware cost.	Only a subset of IOS commands and protocol features is supported.
Safe environment; mistakes cannot damage real equipment.	Simulated behaviour may differ slightly from real devices, so it cannot fully replace hands-on hardware practice.
Easy drag-and-drop interface suitable for beginners.	Not suitable for performance testing, large-scale traffic analysis or security penetration testing.
Simulation mode makes packet flow and OSI layers visible.	Mostly limited to Cisco devices; third-party vendor devices are not available.
Practise anywhere and anytime on a personal computer.	Large topologies with many devices can slow down the application.
Supports both IPv4 and IPv6, wireless and IoT scenarios.	Some features depend on the installed version and need a Networking Academy account to download.
Topologies can be saved, reused and shared with teachers.	Simulates network behaviour but does not teach physical skills such as crimping cables or rack mounting.

5. Applications

- **Education and training.** Used in schools, colleges and universities to teach Computer Networks, from basic LAN setup to routing and switching.
- **Certification preparation.** Lab practice for CCNA, CCNP and other Cisco certification exams.
- **Network design and prototyping.** Testing an IP addressing plan, VLAN layout or routing design before real deployment.
- **Troubleshooting practice.** Finding faults such as wrong IP addresses, incorrect gateways or disconnected links in a prepared scenario.
- **Protocol visualization.** Understanding how ARP, DHCP, DNS, ICMP and routing updates actually work by watching packets in Simulation mode.
- **Internet of Things and automation.** Building smart home or smart city models and writing small scripts to control devices.
- **Competitions and assessments.** Skill contests and graded lab assignments through the Activity Wizard and Multiuser mode.
- **Documentation and demonstrations.** Creating clear topology diagrams for reports, presentations and project work.

Conclusion

Cisco Packet Tracer is a powerful, beginner-friendly tool that brings networking concepts to life. Its mix of a simple interface, real-style configuration, and packet-level visualization makes it ideal for learning and for testing ideas before moving to real equipment. Although it has limits in realism and scale, it remains one of the most effective ways for students to gain practical confidence in computer networks.